



2026 BASIC ABSTRACT ALGEBRA STUDY

LECTURE 1 HOMEWORKS

SOLUTIONS

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Note 1 (Notations)

The following notations will be used throughout the following questions.

- $\mathbb{Z}$: A set of all integers.
- $\mathbb{Z}^+$: A set of all positive integers. i.e., $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$.
- $\mathbb{Z}_n$: A set of non-negative integers from 0 to $n - 1$ for some positive integer n . For instance, $\mathbb{Z}_4 = \{0, 1, 2, 3\}$.
- $n\mathbb{Z}$: A set given as $\{nx \mid x \in \mathbb{Z}\}$ for an integer n . i.e., the set of all multiples of n in $\mathbb{Z}$. For instance, $5\mathbb{Z} = \{\dots, -10, -5, 0, 5, 10, \dots\}$.
- $\mathbb{Q}$: A set of all rational numbers.
- $\mathbb{R}$: A set of all real numbers.
- $\mathbb{R}^+$: A set of all positive real numbers.
- $\mathbb{C}$: A set of all complex numbers.
- $A + B$: A set given as $\{a + b \mid a \in A, b \in B\}$. For instance, $\mathbb{Z}_2 + \mathbb{Z}_3 = \{(0 + 0), (0 + 1), (0 + 2), (1 + 0), (1 + 1), (1 + 2)\} = \{0, 1, 2, 3\}$

Definition 2 (Subset)

Given two sets A and B , A is said to be a subset of B if the following condition holds.

If x is an element of A , then x is an element of B .

Definition 3 (Extensionality)

Two sets A and B are said to be equal if and only if A is a subset of B and B is a subset of A . This is called the *axiom of extensionality*. If two sets A and B are equal, we write $A = B$.

Q1.**EASY**

Which of the following is distinct from the others?

Let A and B be two sets. Let $A(x)$ mean x is an element of A , and $B(x)$ mean x is an element of B .

Which of the following is equivalent to saying that $A \subset B$?

- ☐ 1. $B(x) \rightarrow A(x)$
☐ 2. $\text{not } A(x) \rightarrow B(x)$
☐ 3. $\text{not } A(x) \rightarrow \text{not } B(x)$
- ☐ 4. $A(x) \rightarrow \text{not } B(x)$
☒ 5. $\text{not } B(x) \rightarrow \text{not } A(x)$

A *contrapositive* of a conditional $p \rightarrow q$ is the conditional **not** $q \rightarrow \text{not } p$. A contrapositive is *equivalent* to the original conditional. i.e., it is true if and only if the original conditional is true.

By Definition 2, $A \subset B$ means $A(x) \rightarrow B(x)$ for any x . From the above choices, the only *equivalent* choice is $\text{not } B(x) \rightarrow \text{not } A(x)$, which is a contrapositive of the original conditional.

Therefore, the fifth choice is the correct answer.

Q2.**EASY**

Let Ψ_2 denote the set of all complex solutions to the equation $x^2 = 1$ and let Ψ_4 denote the set of all complex solutions to the equation $x^4 = 1$.

Show that $\Psi_2 \subset \Psi_4$.

Proof. Let's verify that by manually listing elements of Ψ_2 and Ψ_4 .

There are two complex solutions to $x^2 = 1$: 1 and -1 .

Meanwhile, there are four complex solutions to $x^4 = 1$: 1, -1 , i , and $-i$.

From here, clearly $\Psi_2 \subset \Psi_4$. □

Now, let's consider squaring each element of Ψ_2 . By the definition, $1^2 = 1$ and $(-1)^2 = 1$. Again, let's square them once more. $(1^2)^2 = 1^2 = 1$, and similarly $((-1)^2)^2 = 1^2 = 1$. Can you find a pattern?

Q3.

EASY

Prove or disprove that $\mathbb{Z} \subset n\mathbb{Z} + m\mathbb{Z}$ for any two distinct positive integers n and m .

Hint 1. Provide concrete n and m that make up a counterexample. Try justifying your answer using the definition of set builder notation.

The statement is **× false**.

Proof. Consider $n = 10$ and $m = 8$. Take any odd integer, say, 7. For 7 to be an element of $10\mathbb{Z} + 8\mathbb{Z}$, there should be x , a multiple of 10, and y , a multiple of 8, such that $x + y = 7$. Note that every multiple of 10 or 8 is even. As there are no even numbers that become odd when summed, there are no such x and y . Hence, $7 \notin 10\mathbb{Z} + 8\mathbb{Z}$.

As there exists an element $a = 7$ that is an element of $\mathbb{Z}$ but not an element of $10\mathbb{Z} + 8\mathbb{Z}$, the condition $a \in \mathbb{Z} \rightarrow a \in 10\mathbb{Z} + 8\mathbb{Z}$ is false. Thus, $\mathbb{Z} \not\subset 8\mathbb{Z} + 10\mathbb{Z}$. As there is a counterexample in a case where $n = 10$ and $m = 8$, the statement is false. □

Q4.

HARD

Prove or disprove that $2\mathbb{Z} + 3\mathbb{Z} = \mathbb{Z}$.

Hint 1. cf. lecture note.

The statement is ✓ **true**.

Proof. By Definition 3, we need to show that both $2\mathbb{Z} + 3\mathbb{Z} \subset \mathbb{Z}$ and $\mathbb{Z} \subset 2\mathbb{Z} + 3\mathbb{Z}$ hold.

$2\mathbb{Z} + 3\mathbb{Z} \subset \mathbb{Z}$. Recall that $2\mathbb{Z} + 3\mathbb{Z}$ is a set given by $\{2x + 3y \mid x \in \mathbb{Z}, y \in \mathbb{Z}\}$. If n is an element of $2\mathbb{Z} + 3\mathbb{Z}$, then there exist integers x and y such that $n = 2x + 3y$. From here, we conclude that n is an integer, as $2x$ and $3y$ are integers and their sum is also an integer. (cf. PCSA Q2)

$\mathbb{Z} \subset 2\mathbb{Z} + 3\mathbb{Z}$. Let n be an integer. First, let's try to find two integers x and y such that $1 = 2x + 3y$. For instance, there are $x = 2$ and $y = -1$ such that $1 = 2(2) + 3(-1)$. Now, we multiply both sides by n to get $n = n(2x + 3y) = 2xn + 3yn = 2(xn) + 3(yn)$. As n is a sum of a multiple of 2 and a multiple of 3, it is an element of $2\mathbb{Z} + 3\mathbb{Z}$.

By extensionality, we conclude that $2\mathbb{Z} + 3\mathbb{Z} = \mathbb{Z}$. □

In fact, any two coprime integers n and m have the same property. i.e., $n \perp m \rightarrow n\mathbb{Z} + m\mathbb{Z} = \mathbb{Z}$ for integers n, m . Moreover, not only $\mathbb{Z}$ but all principal ideal domains have the same property.

Q5.

HARD

For a positive integer n , let Ψ_n denote the set of all complex solutions to the equation $x^n = 1$.

Show by using Definition 2 that for any two positive integers n and m , $\Psi_n \subset \Psi_{nm}$.

Hint 1. Of course, it is a generalization of Q2.

Recall the Definition 2 that $\Psi_n \subset \Psi_{nm}$ essentially means the following.

If x is a solution to $x^n = 1$, then it is also a solution to $x^{nm} = 1$.

Proof. For two positive integers n and m , assume x is an element of Ψ_n . By the definition, we have $x^n = 1$. By raising both sides to the m th power, we get $(x^n)^m = x^{nm} = 1^m = 1$. From here, we conclude that x is a solution to the equation $x^{nm} = 1$, thus $x \in \Psi_{nm}$.

Therefore, Ψ_n is a subset of Ψ_{nm} . □

Q6.

HARD

Show by Definition 3 that $\mathbb{Z}_n + \mathbb{Z}_n = \mathbb{Z}_{2n-1}$ for any positive integer n .

Hint 1. cf. PCSA Q4.

Or, see the solution of PCSA Q4.

Proof.

$(\mathbb{Z}_n + \mathbb{Z}_n \subset \mathbb{Z}_{2n-1})$. Let $a, b \in \mathbb{Z}_n$. Then $0 \leq a, b \leq n-1$, so $0 \leq a+b \leq 2n-2$, hence $a+b \in \mathbb{Z}_{2n-1}$.

$(\mathbb{Z}_{2n-1} \subset \mathbb{Z}_n + \mathbb{Z}_n)$. Let $k \in \mathbb{Z}_{2n-1}$, so $0 \leq k \leq 2n-2$. If $k \leq n-1$, write $k = k + 0$ with $k, 0 \in \mathbb{Z}_n$. Otherwise $n \leq k \leq 2n-2$, and $k = (n-1) + (k-n+1)$ with both summands in $\{0, 1, \dots, n-1\} = \mathbb{Z}_n$.

By the [Definition 3](#), $\mathbb{Z}_n + \mathbb{Z}_n = \mathbb{Z}_{2n-1}$. □

Q7.

HARD

Prove or disprove that if $\mathbb{Z} = n\mathbb{Z}$ for an integer n , then either $n = 1$ or $n = -1$.

Hint 1. What are the factors of 1 and -1 ?

The statement is ✓ true.

Proof. With the [Definition 3](#) in mind, the assumption says that $\mathbb{Z} \subset n\mathbb{Z}$ and $n\mathbb{Z} \subset \mathbb{Z}$ for some integer n . As the latter is pretty obvious, we will focus on the former.

Recall the definition that $n\mathbb{Z}$ is the set of all multiples of n . And by [Definition 2](#) □

Definition 2 (Symmetric difference)

$A \triangle B$ is a binary set operation given by $(A \setminus B) \cup (B \setminus A)$ and is called a *symmetric difference* between A and B .

Q8.

LEVI

Problem by @escentia07.

Prove or disprove that $A \triangle B \subset C$ if and only if $A \cap B \subset (A \cap B) \cup C$.

The statement is ✓ **true**.

Proof. Forward. Let x be an arbitrary element such that $x \in A \cup B$. We consider two mutually exclusive cases:

(a) Case 1 ($x \in A \cap B$)

If x is in the intersection, it trivially follows that $x \in (A \cap B) \cup C$.

(b) Case 2 ($x \notin A \cap B$)

Since $x \in A \cup B$ but $x \notin A \cap B$, by the definition of symmetric difference, $x \in A \triangle B$. By our assumption $(A \triangle B \subset C)$, this implies $x \in C$. Therefore, $x \in (A \cap B) \cup C$.

Since $x \in (A \cap B) \cup C$ holds true in both cases, we conclude:

$$A \cup B \subset (A \cap B) \cup C$$

Backward. Let x be an arbitrary element such that $x \in A \triangle B$. By definition of the symmetric difference:

$$x \in A \cup B \quad \text{and} \quad x \notin A \cap B$$

By our assumption $(A \cup B \subset (A \cap B) \cup C)$, since $x \in A \cup B$, it must hold that:

$$x \in (A \cap B) \cup C$$

This statement means that $x \in A \cap B$ or $x \in C$. However, we already established that $x \notin A \cap B$. Therefore, it must be true that $x \in C$.

Consequently, we conclude:

$$A \triangle B \subset C$$

□

Q9.

LEVI

For an integer n , let $M(k)$ denote the set $\{x \mid x \equiv k \pmod{n}\}$. Let P the set given by $P = \{M(k) \mid k \in \mathbb{Z}\}$.

- (a)** Show that every two distinct elements of P are disjoint.

Hint 1. Use Definition 3 and *contraposition*.

Proof. Let us rephrase the statement first. The original statement is $M(a) \neq M(b) \rightarrow M(a) \cap M(b) = \emptyset$ for all $a, b \in \mathbb{Z}$. Its contrapositive is $M(a) \cap M(b) \neq \emptyset \rightarrow M(a) = M(b)$ for all $a, b \in \mathbb{Z}$.

Assume $M(a) \cap M(b) \neq \emptyset$ for some $a, b \in \mathbb{Z}$. As it is not empty, there exists at least one element c in $M(a) \cap M(b)$. By the definition, $c \equiv a \pmod{n}$ and $c \equiv b \pmod{n}$.

If x is an element of $M(a)$, then $x \equiv a \equiv c \equiv b \pmod{n}$ by the symmetry and transitivity of the congruence. Hence, x is an element of $M(b)$. Similarly, if x is an element of $M(b)$, then $x \equiv b \equiv c \equiv a \pmod{n}$, showing that $M(a) = M(b)$ as desired. □

If you're up for a challenge, try showing the symmetry and transitivity of the congruence used here.

(b) Show that the following holds for any two integers a and b by [Definition 3](#).

$$M(a) + M(b) = M(a + b)$$

Q10.

LEVI

Problem by @escentia07.

Let $f(x) = x^{2^{121}} + x - 1$

How many real solutions does the equation $f^{2026}(x) = x$ have? Justify your answer.

Answer: 1

As we are to find real solutions, we can think f as a real function. Then, observe that f is strictly increasing, as $x < y$ implies $f(x) < f(y)$.

Now, let S denote the set of all real solutions to the equation $f(x) = x$ and S_{2026} denote the set of all real solutions to the equation $f^{2026}(x) = x$. We will show that $S = S_{2026}$ by the [Definition 3](#).

Proof.

$S \subset S_{2026}$. Let x be an element of S . As $f(x) = x$, we have $f^{2026}(x) = x$, inductively, showing that x is an element of S_{2026} .

$S_{2026} \subset S$. Let x be an element of S_{2026} . If $x < f(x)$, then inductively $x < f^{2026}(x) = x$, which is a contradiction. Similarly, if $x > f(x)$, inductively $x > f^{2026}(x) = x$, which is a contradiction. Therefore, x must be equal to $f(x)$, which amounts to saying that x is an element of S . $\square$

The only real solution to $f(x) = x$ is 1. As S is a singleton, so is S_{2026} .

Therefore, the correct answer is 1.